OTEL-03: Collector Deployment Patterns

> **Series:** OTEL — OpenTelemetry Integration | **Notebook:** 3 of 8 | **Created:** January 2026 | **Last Updated:** 10/29/2026

Deploying the OpenTelemetry Collector

The OTel Collector can be deployed in various patterns depending on your infrastructure. This notebook covers deployment modes, Kubernetes configurations, and best practices for production.

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Prerequisites

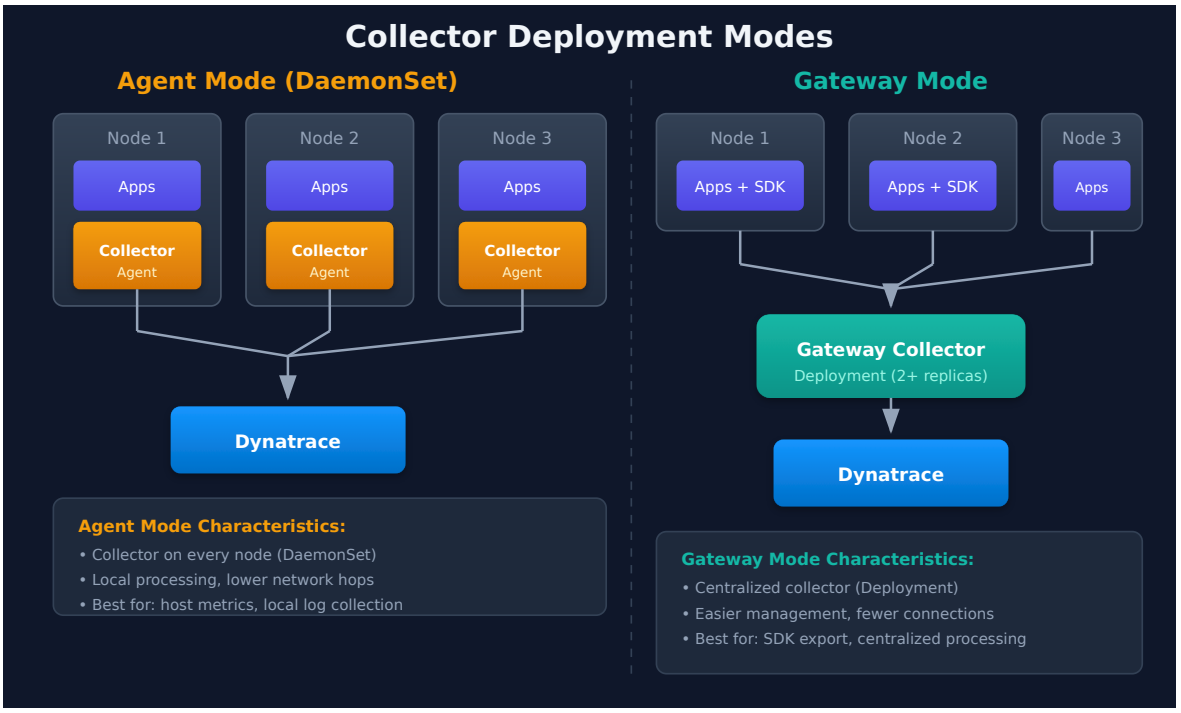
Requirement	Details
Knowledge	OTEL-02: Collector Architecture
Tools	kubectl, Helm, or Docker

1. Deployment Modes

Mode Comparison

Mode	Deployment	Use Case	Pros	Cons
Agent	DaemonSet/Sidecar	Per-node collection	Low latency, local processing	More instances
Gateway	Deployment	Centralized	Single point, aggregation	Potential bottleneck
Hybrid	Both	Large scale	Best of both	Complex

Architecture Patterns



2. Agent Mode (Sidecar/DaemonSet)

DaemonSet Configuration

> **Important:** Always pin your Collector image to a specific version. Using `latest` can cause unexpected behavior during upgrades. Check the OpenTelemetry Collector releases for the current stable version.

```yaml
apiVersion: apps/v1
kind: DaemonSet
metadata:
  name: otel-collector-agent
  namespace: otel
spec:
  selector:
    matchLabels:
      app: otel-collector-agent
  template:
    metadata:
      labels:
        app: otel-collector-agent
    spec:
      containers:
        - name: collector
          image: otel/opentelemetry-collector-contrib:0.151.0
          args:
            - --config=/conf/otel-collector-config.yaml
          ports:
            - containerPort: 4317  # OTLP gRPC
            - containerPort: 4318  # OTLP HTTP
          resources:
            requests:
              cpu: 100m
              memory: 256Mi
            limits:
              cpu: 500m
              memory: 512Mi
          volumeMounts:
            - name: config
              mountPath: /conf
      volumes:
        - name: config
          configMap:
            name: otel-collector-config
```

Sidecar Pattern

```
apiVersion: v1
kind: Pod
metadata:
  name: my-app
spec:
  containers:
    - name: app
      image: my-app:latest
      env:
        - name: OTEL_EXPORTER_OTLP_ENDPOINT
          value: http://localhost:4317
    - name: otel-collector
      image: otel/opentelemetry-collector-contrib:0.151.0
      args:
        - --config=/conf/otel-collector-config.yaml
      ports:
        - containerPort: 4317
```

> **Collector version currency (checked 10/28/2026):** the pins in this notebook (`otel/opentelemetry-collector-contrib:0.151.0` , Dynatrace distro `v0.48.0`) are the **validated baseline** for these examples. Newer Dynatrace-distro releases exist — up to **v0.59.0** (released 10/27/2026, wrapping upstream v0.163.0). Review the release notes for every version you cross before bumping; earlier examples of breaking changes include the `filter` processor's default `error_mode` changing to `ignore` (0.49), `span_metrics` rejecting duplicate dimension names at startup (0.50), and `memory_limiter` self-metrics being renamed to `otelcol_processor_memory_limiter_*` (0.51 — affects self-monitoring dashboards that chart those series). Upgrade deliberately per the always-pin guidance in this section, reviewing the release notes for each version you cross.
>
> **Production image:** the DaemonSet, sidecar and gateway examples pin Contrib so the YAML runs anywhere. For production, swap in the pinned Dynatrace distribution (`ghcr.io/dynatrace/dynatrace-otel-collector/dynatrace-otel-collector:v0.48.0`) — Dynatrace *"recommends using the Contrib distribution only for testing purposes"*. OTEL-07 § 1 covers what Dynatrace Support covers.

3. Gateway Mode

Gateway Deployment

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: otel-collector-gateway
  namespace: otel
spec:
  replicas: 2  # HA
  selector:
    matchLabels:
      app: otel-collector-gateway
  template:
    metadata:
      labels:
        app: otel-collector-gateway
    spec:
      containers:
        - name: collector
          image: otel/opentelemetry-collector-contrib:0.151.0  # pinned — see
the version-pinning callout in §2
          args:
            - --config=/conf/otel-collector-config.yaml
          ports:
            - containerPort: 4317
            - containerPort: 4318
          resources:
            requests:
              cpu: 500m
              memory: 1Gi
            limits:
              cpu: 2000m
              memory: 4Gi
```

Gateway Service

```
apiVersion: v1
kind: Service
metadata:
  name: otel-collector-gateway
  namespace: otel
spec:
  ports:
    - name: otlp-grpc
      port: 4317
      targetPort: 4317
    - name: otlp-http
      port: 4318
      targetPort: 4318
  selector:
    app: otel-collector-gateway
```

4. Kubernetes Deployment

Helm Installation

```
# Add Helm repo
helm repo add open-telemetry https://open-telemetry.github.io/opentelemetry-helm-charts
helm repo update

# Install as DaemonSet (agent mode)
helm install otel-collector open-telemetry/opentelemetry-collector \
  --namespace otel \
  --create-namespace \
  --set mode=daemonset \
  --set image.repository="ghcr.io/open-telemetry/opentelemetry-collector-releases/opentelemetry-collector-k8s" \
  --set command.name="otelcol-k8s"

# Install as Deployment (gateway mode)
helm install otel-collector open-telemetry/opentelemetry-collector \
  --namespace otel \
  --create-namespace \
  --set mode=deployment \
  --set replicaCount=2 \
  --set image.repository="ghcr.io/open-telemetry/opentelemetry-collector-releases/opentelemetry-collector-k8s" \
  --set command.name="otelcol-k8s"
```

The chart ships with no default image, so `image.repository` must be set or the install stops with an error; `command.name` must match the image (`otelcol-k8s` for the `-k8s` distribution).

> **Sources:** opentelemetry-collector chart NOTES.txt (OpenTelemetry GitHub) — *"'image.repository' must be set."*; opentelemetry-collector chart README (OpenTelemetry GitHub) — install line with `--set image.repository=… --set command.name="otelcol-k8s"`.

Helm Values for Dynatrace

```
# values.yaml
mode: deployment
replicaCount: 2

image:
  repository: ghcr.io/open-telemetry/opentelemetry-collector-
releases/opentelemetry-collector-k8s
command:
  name: otelcol-k8s

config:
  receivers:
    otlp:
      protocols:
        grpc:
          endpoint: 0.0.0.0:4317
        http:
          endpoint: 0.0.0.0:4318

  processors:
    batch:
      timeout: 10s
    memory_limiter:
      check_interval: 1s
      limit_mib: 800

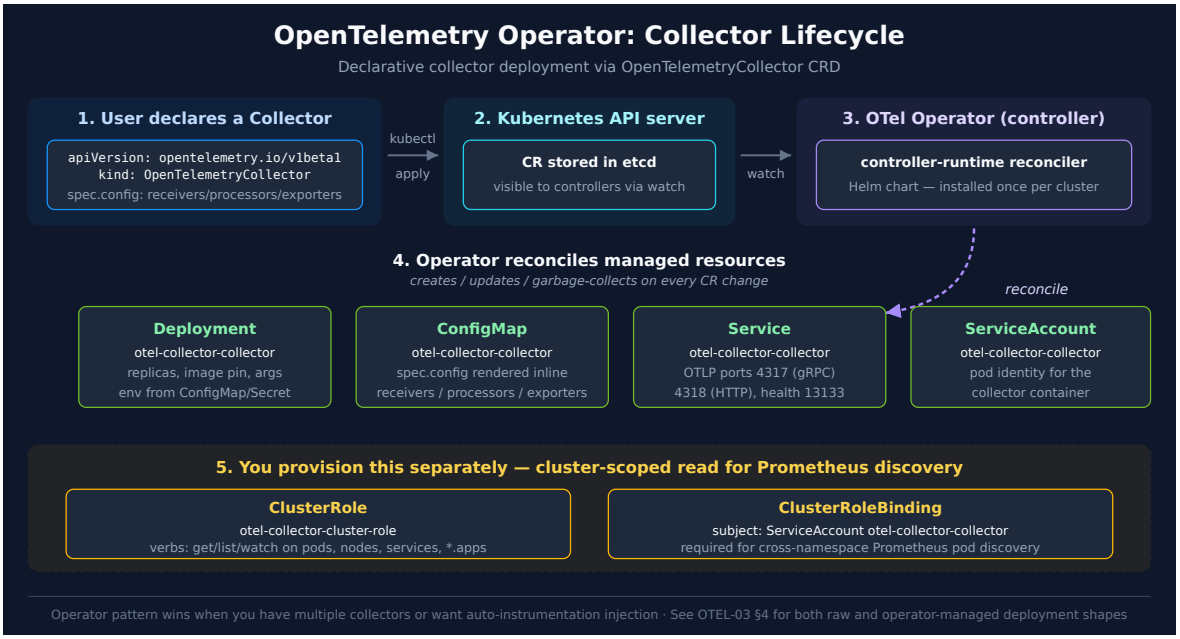
  exporters:
    otlphttp:
      endpoint: https://${DT_ENDPOINT}/api/v2/otlp
      headers:
        Authorization: Api-Token ${DT_TOKEN}

  service:
    pipelines:
      traces:
        receivers: [otlp]
        processors: [memory_limiter, batch]
        exporters: [otlphttp]
      metrics:
        receivers: [otlp]
        processors: [memory_limiter, batch]
        exporters: [otlphttp]
      logs:
        receivers: [otlp]
```

```
        processors: [memory_limiter, batch]
        exporters: [otlphttp]
```

OpenTelemetry Operator (CRD-managed)

The Helm install above creates a raw `Deployment` that you own and roll over manually. The **OpenTelemetry Operator** introduces a higher-level pattern — install the operator once, then declare collectors as `OpenTelemetryCollector` custom resources. The operator reconciles the `Deployment`, `ConfigMap`, `ServiceAccount`, and the collector's container.



Two-stage install:

```
# Stage 1 — install the operator (once per cluster)
helm repo add open-telemetry https://open-telemetry.github.io/opentelemetry-
helm-charts
helm repo update

helm install opentelemetry-operator open-telemetry/opentelemetry-operator \
  --namespace opentelemetry-operator-system \
  --create-namespace \
  --set "manager.collectorImage.repository=otel/opentelemetry-collector-k8s" \
  --set admissionWebhooks.certManager.enabled=false \
  --set admissionWebhooks.autoGenerateCert.enabled=true \
  --wait --timeout 5m

# Stage 2 — declare a collector via custom resource
kubectl apply -f opentelemetry-collector.yaml
```

`OpenTelemetryCollector` resource shape:

```
apiVersion: opentelemetry.io/v1beta1
kind: OpenTelemetryCollector
metadata:
  name: otel-collector
  namespace: o11y
spec:
  env:
    - name: DT_ENDPOINT
      valueFrom:
        configMapKeyRef:
          name: otel-collector-config
          key: DT_ENDPOINT
    - name: DT_API_TOKEN
      valueFrom:
        secretKeyRef:
          name: otel-collector-secret
          key: DT_API_TOKEN
  config:
    receivers:
      prometheus:
        config:
          scrape_configs:
            - job_name: kubernetes-pods
              # See OTEL-05 §6 for the full receiver + relabel chain
    processors:
      memory_limiter:
        check_interval: 1s
        limit_percentage: 75
      batch:
        send_batch_size: 10000
        timeout: 10s
    exporters:
      otlphttp:
        endpoint: ${env:DT_ENDPOINT}
        headers:
          Authorization: Api-Token ${env:DT_API_TOKEN}
    service:
      pipelines:
        metrics:
          receivers: [prometheus]
          processors: [memory_limiter, batch]
          exporters: [otlphttp]
```

The operator creates the `ServiceAccount` automatically (named `<collectorName>-collector`). For Prometheus pod discovery the collector still needs cluster-scoped read

access — provision a `ClusterRole` and bind it separately:

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: otel-collector-cluster-role
rules:
  - apiGroups: ["", "apps", "batch"]
    resources:
      - pods
      - namespaces
      - nodes
      - services
      - replicasets
      - deployments
      - daemonsets
      - statefulsets
      - jobs
      - cronjobs
    verbs: ["get", "list", "watch"]
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: otel-collector-cluster-role-binding
subjects:
  - kind: ServiceAccount
    name: otel-collector-collector   # &lt;collectorName&gt;-collector
    namespace: o11y
roleRef:
  kind: ClusterRole
  name: otel-collector-cluster-role
  apiGroup: rbac.authorization.k8s.io
```

When to use which — in community practice, the choice tends to follow collector count and whether you want the operator's extras; weigh it against how your team already runs rollouts:

Choose	When
Raw `Deployment` + Helm chart	One or two collectors; team already manages Deployment rollouts; Helm-first workflows
`OpenTelemetryCollector` CRD + Operator	Multiple collectors per cluster; auto-reload on config change; want OTLP auto-instrumentation injection for application pods

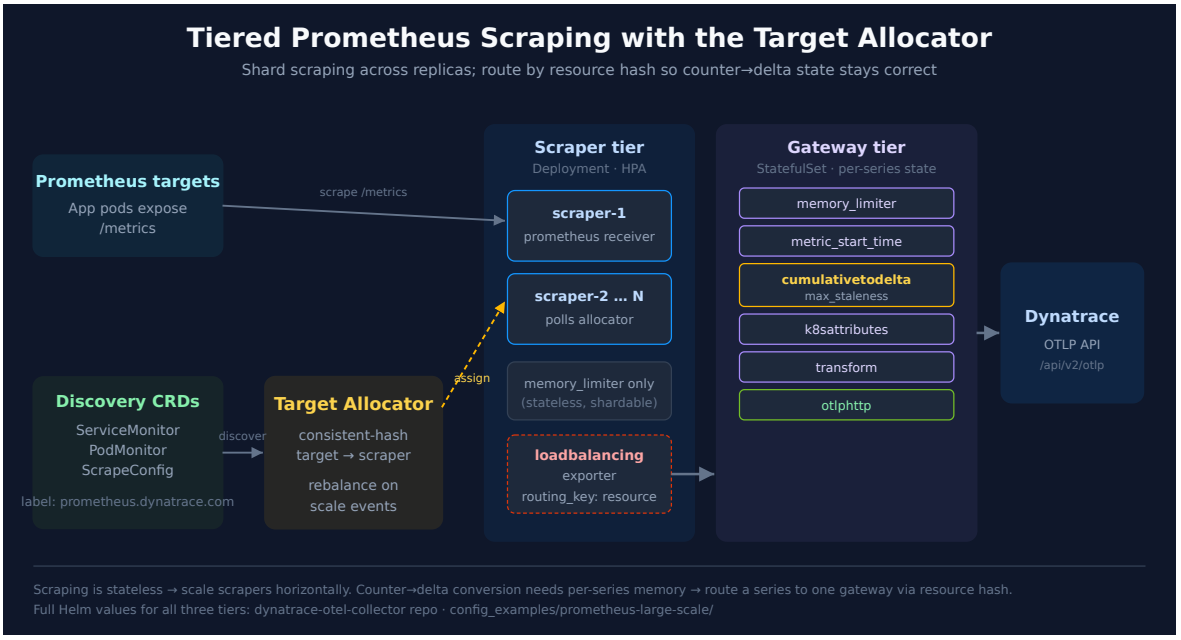
In community practice, the operator's auto-instrumentation feature is often the deciding factor for greenfield Kubernetes deployments — annotate a pod with `instrumentation.opentelemetry.io/inject-python: "true"` and the operator injects an init container that loads the OTel agent, pointed at a `Service` your collector exposes. See **OTEL-04 — Trace Instrumentation** for the application-side pattern.

> **Sources:**
>
> - OpenTelemetry Operator (opentelemetry-operator GitHub)
> - OpenTelemetryCollector CRD reference (opentelemetry-operator GitHub)
> - Auto-instrumentation (opentelemetry-operator GitHub) — *"The operator can inject and configure OpenTelemetry auto-instrumentation libraries."*

5. Scaling Prometheus Scraping: Target Allocator & Tiered Collectors

The single-collector pattern in **OTEL-05 — Metrics Instrumentation, §6** (one collector with a `prometheus` receiver scraping annotated pods) is the right starting point for a small or static set of endpoints. But it does not scale horizontally: a `prometheus` receiver owns a fixed target list, so a second replica makes **both** collectors scrape **every** target — double-counting metrics, not sharing load. Once you reach thousands of scrape targets or millions of data points per minute, you need a tiered architecture that shards scraping across replicas.

This builds directly on the OpenTelemetry Operator pattern in §4 — the **Target Allocator** is an Operator component.



Why two tiers

Concern	Tier	Why it must live here
Target discovery & sharding	Target Allocator	One brain assigns targets to scrapers and rebalances on scale events
Scraping	Scraper (Deployment)	Stateless and shardable — scale horizontally with an HPA
Counter→delta conversion, enrichment	Gateway (StatefulSet)	`cumulativetodelta` keeps **per-series memory state**; every sample for a series must reach the **same** instance

The split exists because of `cumulativetodelta`. Prometheus counters are cumulative; Dynatrace requires delta temporality (cross-reference **OTEL-07 — Dynatrace Integration, §9**). Converting cumulative→delta requires remembering the previous value **per series**, so it cannot be sharded blindly — the gateway tier receives a **resource-hashed** stream so all samples of a series land on one gateway.

Target Allocator

The Allocator discovers targets from **Prometheus Operator CRDs** and distributes them across scrapers by consistent hashing. Each scraper's `prometheus` receiver polls the Allocator for its assigned slice instead of holding a static `scrape_configs` list:

```
# Scraper tier — prometheus receiver driven by the Target Allocator
receivers:
  prometheus:
    target_allocator:
      endpoint: http://tiered-allocator-ta
      interval: 15s
      collector_id: ${env:K8S_POD_NAME}   # unique per scraper pod
```

The Allocator selects which CRDs to honor with label selectors:

```
# Target Allocator config
prometheus_cr:
  enabled: true
  scrapeInterval: 15s
  service_monitor_selector:
    matchlabels:
      prometheus.dynatrace.com: "true"
  pod_monitor_selector:
    matchlabels:
      prometheus.dynatrace.com: "true"
  scrape_config_selector:
    matchlabels:
      prometheus.dynatrace.com: "true"
```

Targets are declared with the same three CRDs a Prometheus Operator stack uses — opt them in with the matching label:

```
apiVersion: monitoring.coreos.com/v1
kind: ServiceMonitor
metadata:
  name: my-service
  labels:
    prometheus.dynatrace.com: "true"
spec:
  selector:
    matchLabels:
      app: my-service
  endpoints:
    - port: metrics
      interval: 60s
```

`PodMonitor` selects pods directly (bypassing Services); `ScrapeConfig` is the native Prometheus format for annotation-based discovery.

Scraper tier → gateway tier

Scrapers do **no** stateful processing — a `memory_limiter` and then hand off via the **`loadbalancing`** exporter, keyed by `resource` so all series from a given source resource route to one gateway:

```
# Scraper tier — route to gateways by resource hash
exporters:
  loadbalancing:
    routing_key: resource          # load-bearing: keeps a series on one
gateway
    resolver:
      k8s:
        service: tiered-gateway.otel-ta
    protocol:
      otlp:
        tls:
          insecure: true
```

The gateway tier does the stateful work — converting counters, enriching with Kubernetes attributes, exporting to Dynatrace:

```
# Gateway tier (StatefulSet)
processors:
  memory_limiter:
    check_interval: 1s
    limit_percentage: 95
    spike_limit_percentage: 5
  metric_start_time:
  cumulativetodelta:
    max_staleness: 10m        # ~10x scrape interval; evicts stale series
    initial_value: drop
  k8sattributes:
    # extract metadata.dynatrace.com/* annotations + k8s.* resource attributes
  transform: {}

exporters:
  otlphttp:
    endpoint: ${env:DT_ENDPOINT}
    headers:
      Authorization: "Api-Token ${env:DT_API_TOKEN}"

service:
  pipelines:
    metrics:
      receivers: [otlp]
      processors: [memory_limiter, metric_start_time, cumulativetodelta,
k8sattributes, transform]
      exporters: [otlphttp]
```

`max_staleness` bounds gateway memory: a series not seen within that window is evicted, so pod churn does not grow state unbounded. Set it to roughly 10x the scrape interval.

Scaling the fleet

Both tiers run an HPA. Scale **up** immediately and **down** slowly, so a brief lull does not drop a replica mid-series and reset delta state:

```
behavior:
  scaleUp:
    stabilizationWindowSeconds: 0
  scaleDown:
    stabilizationWindowSeconds: 300   # ~5x scrape interval
```

Replica counts scale roughly with data-point throughput. Dynatrace publishes a reference sizing (60-second scrape interval) — treat these as a **starting point and size against**

your own data-points-per-minute and target count, not as fixed values:

Data points/min	Targets	Scraper replicas	Gateway replicas
1M	~5	2	2
10M	~60	13	4
30M	~334	38	8
60M	~667	74	18
90M	~1,000	102	23

Hash-based distribution does not rebalance on actual load — a heavy target stays pinned to its assigned scraper. If one target dominates, split it across multiple ports/paths or vertically size the scraper that owns it.

Deploying the tiers

Four components deploy in the order the reference repo prescribes — RBAC and discovery CRs first, then one Helm release per tier:

```
# 1. ServiceAccounts, ClusterRoles, bindings + the ScrapeConfig CRs
kubectl apply -f rbac.yaml
kubectl apply -f scrapeconfig.yaml

# 2. one Helm release per component
helm upgrade --install allocator       &lt;chart&gt; -f allocator.values.yaml
helm upgrade --install tier1-scraper   &lt;chart&gt; -f tier1-
scraper.values.yaml
helm upgrade --install tier2-gateway   &lt;chart&gt; -f tier2-
gateway.values.yaml
helm upgrade --install selfmon-scraper &lt;chart&gt; -f selfmon-scraper.yaml
# recommended
```

The repo ships one values file per component (`allocator.values.yaml` , `tier1-scraper.values.yaml` , `tier2-gateway.values.yaml` , `selfmon-scraper.yaml`) plus `rbac.yaml` and `scrapeconfig.yaml` . **Prerequisite:** the Prometheus Operator CRDs (`ServiceMonitor` / `PodMonitor` / `ScrapeConfig`) must already be installed — the Allocator consumes them.

RBAC: what each tier needs

Each tier runs under its own `ServiceAccount` bound to a `ClusterRole` scoped to only what that tier does — the Allocator needs the broadest access (it does discovery), scrapers need target-resolution reads, gateways need only the metadata `k8sattributes` enriches with:

ServiceAccount	Reads	Why
`tiered-otel-allocator`	pods, endpoints, services, endpointslices, nodes, nodes/metrics, configmaps, secrets, ingresses, namespaces + Prometheus CRDs (`servicemonitors` , `podmonitors` , `scrapeconfigs` , `probes`) + non-resource URL `/metrics`	Discovers targets from CRDs and the API server
`tiered-otel-scraper`	pods, endpoints, services, endpointslices, nodes, namespaces, replicasets, deployments, statefulsets, daemonsets, jobs, cronjobs (`get` / `list` / `watch`)	Resolves scrape targets behind Services/Endpoints
`tiered-otel-gateway`	pods, namespaces, nodes, replicasets, jobs, cronjobs (`get` / `list` / `watch`)	`k8sattributes` enrichment metadata only

Worked `ClusterRole` for the Allocator — the load-bearing one. Note the Prometheus CRD verbs and the `/metrics` non-resource URL, both of which the scraper/gateway roles omit:

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: tiered-otel-allocator
rules:
  - apiGroups: [""]
    resources: [pods, endpoints, services, nodes, nodes/metrics, configmaps,
secrets, namespaces]
    verbs: [get, list, watch]
  - apiGroups: ["discovery.k8s.io"]
    resources: [endpointslices]
    verbs: [get, list, watch]
  - apiGroups: ["monitoring.coreos.com"]
    resources: [servicemonitors, podmonitors, scrapeconfigs, probes]
    verbs: ["*"]
  - nonResourceURLs: ["/metrics"]
    verbs: [get]
---
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: tiered-otel-allocator
subjects:
  - kind: ServiceAccount
    name: tiered-otel-allocator
    namespace: otel-ta
roleRef:
  kind: ClusterRole
  name: tiered-otel-allocator
  apiGroup: rbac.authorization.k8s.io
```

Each of the four ServiceAccounts gets a matching `ClusterRoleBinding`. Unlike the `OpenTelemetryCollector` CRD path in §4 (where the Operator creates the collector ServiceAccount for you), this Helm-values deployment provisions all four explicitly in `rbac.yaml`.

Target Allocator high availability

The Allocator sits on the control path — if it is unreachable, scrapers cannot learn their assignments. Run **multiple Allocator replicas**. Because the assignment algorithm is deterministic, the replicas need no leader election or shared state:

> "Run multiple TA replicas for failure tolerance. Because the consistent-hashing algorithm is deterministic, all replicas independently produce the same target-to-

> scraper assignments without coordination."

Every replica computes the identical target→scraper map from the same discovered set, so a scraper polling any replica gets a consistent answer and losing a replica is transparent — the scraper simply polls another.

Self-monitoring & verification

The reference deployment ships a dedicated **self-monitoring collector** (`selfmon-scraper`) whose only job is to scrape the other collectors' and the allocator's internal telemetry and forward it to Dynatrace — the fleet observing itself. It is a `prometheus` receiver doing pod discovery in the collector namespace, keyed off annotations:

```
receivers:
  prometheus:
    config:
      scrape_configs:
        - job_name: otel-selfmon
          scrape_interval: 15s
          kubernetes_sd_configs:
            - role: pod
              namespaces:
                names: [otel-ta]
          # keep pods annotated metrics.dynatrace.com/scrape: "true";
          # read the scrape port from metrics.dynatrace.com/port
exporters:
  otlphttp/dynatrace:
    endpoint: ${env:DT_ENDPOINT}
    headers:
      Authorization: "Api-Token ${env:DT_API_TOKEN}"
```

Watch these collector self-metrics — they tell you whether the fleet is keeping up and whether routing is intact:

Metric	Tier	Healthy signal
`otelcol_receiver_accepted_metric_points`	Scraper	Rises in step with scrape volume; flat-lining means scraping stalled

Metric	Tier	Healthy signal
`otelcol_processor_incoming_items` vs `otelcol_processor_outgoing_items`	Gateway	Roughly equal; a growing gap means a processor (usually `memory_limiter`) is dropping
`otelcol_exporter_send_failed_metric_points`	Gateway	Near zero; sustained non-zero means export to Dynatrace is failing (auth, endpoint, or rate)
`otelcol_loadbalancer_num_resolutions`	Scraper	Changes when the gateway set scales; a value stuck through a known scale event means stale routing

Once these land in Dynatrace you can chart them with `timeseries` like any other metric and alert on the export-failure and processor-gap signals. Confirm the exact ingested metric keys in the metric browser first — collector self-metric naming varies by distribution and OTel Collector version.

Single collector vs. tiered — which to run

Run	When
Single collector (OTEL-05 §6)	A small or static set of endpoints; no need for auto-scaling or redundancy
Tiered + Target Allocator	Thousands of pods or hundreds of distinct endpoints; millions of data points per minute; HPA-driven horizontal scaling instead of manually partitioning targets; CRD-based (`ServiceMonitor` / `PodMonitor` / `ScrapeConfig`) discovery

Complete, runnable configs (Helm values for all three tiers, RBAC, self-monitoring) live in the Dynatrace OTel Collector repo under `config_examples/prometheus-large-scale/`.

> **Sources:**
>
> - Prometheus standard use case (DT docs) — *"The metric_start_time and cumulative_to_delta*

> *processors are stateful and must see every consecutive sample of a series in one place to compute deltas correctly."*
>
> - prometheus-large-scale config examples (Dynatrace GitHub) — tier-1 scraper / tier-2 gateway / allocator Helm values, adapted to the v0.151.0 pin (`cumulativetodelta` ; the repo now spells it `cumulative_to_delta`)
> - OpenTelemetry Operator (opentelemetry-operator GitHub) — Target Allocator component
> - About OTLP metrics ingest (DT docs) — *"The Dynatrace backend exclusively works with delta values and requires the respective aggregation temporality."*

6. Docker Compose

Basic Setup

```
# docker-compose.yaml
version: '3.8'

services:
  otel-collector:
    image: otel/opentelemetry-collector-contrib:0.151.0  # pinned — never
latest, even in local/dev compose files
    command: ["--config=/etc/otel-collector-config.yaml"]
    volumes:
      - ./otel-collector-config.yaml:/etc/otel-collector-config.yaml
    ports:
      - "4317:4317"   # OTLP gRPC
      - "4318:4318"   # OTLP HTTP
      - "13133:13133" # Health check
    environment:
      - DT_ENDPOINT=${DT_ENDPOINT}
      - DT_TOKEN=${DT_TOKEN}

  my-app:
    image: my-app:latest
    environment:
      - OTEL_EXPORTER_OTLP_ENDPOINT=http://otel-collector:4317
      - OTEL_SERVICE_NAME=my-app
    depends_on:
      - otel-collector
```

7. High Availability

HA Considerations

Component	HA Strategy
Agent (DaemonSet)	Node failure = pod restarts
Gateway	Multiple replicas behind load balancer
Persistent Queue	Prevent data loss during restarts

Gateway with Persistent Queue

```
exporters:
  otlphttp:
    endpoint: https://${DT_ENDPOINT}/api/v2/otlp
    sending_queue:
      enabled: true
      num_consumers: 10
      queue_size: 1000
      storage: file_storage
    retry_on_failure:
      enabled: true
      initial_interval: 1s
      max_interval: 30s
      max_elapsed_time: 300s

extensions:
  file_storage:
    directory: /var/lib/otel/storage
    timeout: 10s

service:
  extensions: [file_storage]
```

Pod Disruption Budget

```
apiVersion: policy/v1
kind: PodDisruptionBudget
metadata:
  name: otel-collector-pdb
spec:
  minAvailable: 1
  selector:
    matchLabels:
      app: otel-collector-gateway
```

8. Resource Sizing

Sizing Guidelines

These are community starting points, not published limits — measure your own spans per second and memory headroom, and resize from there:

Throughput	CPU	Memory	Replicas
Low (<1000 spans/s)	100m	256Mi	1
Medium (<10k spans/s)	500m	1Gi	2
High (<100k spans/s)	2	4Gi	3+
Very High (>100k spans/s)	4+	8Gi+	5+

Memory Limiter Tuning

```
processors:
  memory_limiter:
    check_interval: 1s
    limit_mib: 800       # 80% of container memory limit
    spike_limit_mib: 200 # Room for spikes
```

Horizontal Pod Autoscaler

```
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: otel-collector-hpa
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: otel-collector-gateway
  minReplicas: 2
  maxReplicas: 10
  metrics:
    - type: Resource
      resource:
        name: cpu
        target:
          type: Utilization
          averageUtilization: 70
```

9. Security Considerations

> **Workload Identity Federation — credential-free auth (SaaS 1.346 — staged rollout from 08/25/2026).** Verbatim: *"Workload Identity Federation (WIF) enables secure, credential-free authentication when exporting telemetry to external services and APIs. It removes the need for long-lived credentials in Dynatrace OpenTelemetry Collector deployments, following modern cloud-native security practices."* Where it applies, this removes the long-lived API token from the Collector's configuration and secret store entirely — the failure mode this section otherwise has to mitigate with rotation and secret management. Treat the token-based patterns shown here as the working path until WIF reaches your tenant and you have confirmed your export targets support it.
>
> Source: What's new in Dynatrace SaaS 1.346 (DT docs)

Token Management

Store Dynatrace API tokens in Kubernetes Secrets:

```
# Kubernetes Secret (create via kubectl or sealed-secrets)
apiVersion: v1
kind: Secret
metadata:
  name: dynatrace-otel-token
type: Opaque
stringData:
  token: &lt;your-dynatrace-api-token&gt;
```

```
# Reference in Deployment
env:
  - name: DT_TOKEN
    valueFrom:
      secretKeyRef:
        name: dynatrace-otel-token
        key: token
```

TLS Configuration

```
receivers:
  otlp:
    protocols:
      grpc:
        endpoint: 0.0.0.0:4317
        tls:
          cert_file: /certs/server.crt
          key_file: /certs/server.key
          client_ca_file: /certs/ca.crt  # mTLS
```

Network Policy

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: otel-collector-policy
spec:
  podSelector:
    matchLabels:
      app: otel-collector-gateway
  policyTypes:
    - Ingress
    - Egress
  ingress:
    - from:
        - namespaceSelector: {}
      ports:
        - protocol: TCP
          port: 4317
        - protocol: TCP
          port: 4318
  egress:
    - to:
        - ipBlock:
            cidr: 0.0.0.0/0
      ports:
        - protocol: TCP
          port: 443
```

Summary

In this notebook, you learned:

- Deployment modes: Agent vs. Gateway vs. Hybrid
- DaemonSet and Sidecar configurations for agent mode
- Gateway deployment with services
- Kubernetes deployment via Helm
- Docker Compose setup
- High availability with persistent queues
- Resource sizing guidelines
- Security best practices

Next Steps

Next Notebook	Topic
OTEL-04: Trace Instrumentation	Instrumenting code
OTEL-07: Dynatrace Integration	Complete DT setup

References

- Collector Deployment
- Helm Chart
- Docker Images
- Dynatrace OTel Collector (DT docs) — *"Dynatrace recommends using the Contrib distribution only for testing purposes."*

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.